

Tokenized real-world asset

Real-World Asset or **RWA** is an asset class within blockchain and cryptocurrency that represents a digital token of a tangible or intangible asset existing outside the blockchain ecosystem.^[1] These tokens are created through a process called tokenization, which aims to bridge the gap between traditional finance (TradFi) and decentralized finance (DeFi).^[2] RWAs can represent a wide range of assets, such as real estate, government and corporate bonds, commodities like gold, private credit, and works of art.^{[1][3]} This asset class is considered a key driver of digital asset adoption, with some financial leaders, such as BlackRock CEO Larry Fink, describing the underlying technology as "the next generation for markets."^[4] The asset class has garnered significant attention from financial regulators, with bodies such as the U.S. President's Working Group on Digital Asset Markets outlining frameworks for its integration into national financial systems.^[5]

Creation through Tokenization

The creation of an RWA token involves linking an off-chain asset to its on-chain digital equivalent through a process known as tokenization. This typically involves three stages:^[2]

1. **Off-chain Formalization:** This initial stage involves confirming the value and legal ownership of the physical asset. It includes processes like property valuation, legal registration, and establishing clear documentation that links the asset to the token that will be created.
2. **Information Bridging:** Data regarding the asset's value, ownership, and legal status is converted into digital metadata. This information is often made accessible to the blockchain through a blockchain oracle, which is a secure service that provides external data to smart contracts. Few established decentralized oracle networks that supply real-time price and asset data include Chainlink, Pyth Network (<https://www.pyth.network/>), Band Protocol (<https://www.bandprotocol.com/>), and API3 (<https://api3.org/>).^[6]
3. **Token Minting and Distribution:** A smart contract is used to mint (create) the digital tokens representing the asset. These tokens are then issued on the blockchain, and their ownership can be transferred, traded, or used within DeFi applications.

Characteristics and Benefits

The primary purpose of RWAs is to bring the value and utility of off-chain assets into the on-chain digital economy. Proponents argue that this asset class offers several advantages:^[3]

Increased Liquidity

RWAs can transform traditionally illiquid assets, such as real estate or fine art, into tradable digital tokens. This allows them to be bought and sold more easily on secondary markets, increasing their liquidity.^[1]

Fractional Ownership

High-value assets can be divided into smaller, more affordable tokens representing fractional ownership. This lowers the barrier to entry for smaller investors to own a share

of assets that would otherwise be inaccessible, such as commercial real estate or rare collectibles.^[2]

Greater Accessibility and Efficiency

As a digital asset class, RWAs can democratize access to global investment opportunities. Blockchain technology can also streamline transactions, enabling faster settlement times and reduced administrative overhead compared to traditional financial systems.^[3]

Transparency

The ownership records and transaction histories of RWA tokens are immutably recorded on the blockchain which provides a high degree of transparency for all participants.^[1]

Common examples of RWAs include tokenized government securities (like U.S. Treasury bills), real estate properties, private credit instruments, and corporate invoices.^[2]

Challenges and Risks

Despite its potential, the RWA asset class faces several significant challenges:

Regulatory Uncertainty

The legal and regulatory frameworks for tokenized assets are still developing in many jurisdictions. Issues related to securities law, property rights, and Know Your Customer (KYC) / Anti-Money Laundering (AML) compliance remain complex hurdles.^[2]

Asset Custody and Valuation

Ensuring the secure custody of the underlying physical asset and accurately reflecting its real-time value on the blockchain is a major operational challenge. The reliability of the valuation process is critical for maintaining trust in the token's worth.^[1]

Oracle Reliability

The system's dependence on oracles to feed real-world data (like asset prices) to the blockchain introduces a potential point of failure. If an oracle is compromised or provides inaccurate data, it can have significant consequences for the associated smart contracts.^[3]

Technical and Security Risks

As with any blockchain-based system, there are inherent risks related to smart contract vulnerabilities, platform security, and potential cyberattacks.

Market Size and Adoption

The RWA asset class is a rapidly growing sector within the cryptocurrency industry. Financial institutions and consulting firms have projected significant growth for the market. A 2022 report by the Boston Consulting Group forecasted that the market for tokenized illiquid assets could become a US\$16 trillion business opportunity by 2030.^[7] Similarly, a report by Bank of America in 2023 highlighted RWA tokenization as a "key driver of digital asset adoption."^[8]

Several prominent blockchain projects and financial companies are active in the RWA space, including *Centrifuge*, a platform for tokenizing assets like invoices, Securitize and *Ondo Finance*, which offer tokenized U.S. bonds and stocks, and *Mantra Chain*, a company focusing on the tokenization of real estate, particularly in the UAE through a partnership agreement with Damac Properties.^{[3][2][9]} In February 2026, the cryptocurrency wallet MetaMask integrated services from *Ondo Finance*, allowing eligible mobile users in supported non-U.S. jurisdictions to access tokenized U.S. financial assets within the application. The integration enabled trading of more than 200 tokenized securities.^[10]

Notable Implementations

As the RWA sector matured into 2025, several protocols and companies established significant market presence by tokenizing specific asset classes:

Financial Instruments and Securities

Ondo Finance and **Securitize** became leaders in the tokenization of U.S. Treasuries and money market funds. Ondo's OUSG (tokenized short-term government bonds) and Securitize's collaboration with institutional managers aimed to provide "risk-free" yield to on-chain investors by bridging regulated financial instruments into DeFi ecosystems.^[11]

Real Estate

Mantra Chain specialized in the tokenization of high-value real estate. In early 2025, the platform signed a landmark US\$1 billion agreement with the Dubai-based **DAMAC Group** to tokenize a portfolio of luxury properties, hospitality assets, and data centers, allowing for fractional ownership of Middle Eastern real estate on a public ledger.^[12]

Rare Commodities and Gemstones

In 2024, the Australian gemstone firm **NYBlue Pty Ltd** introduced the **ZIRC token**, a niche RWA designed to establish a resource-backed market for Cambodian blue zircon. Each token is purportedly backed by and redeemable for one carat of physical blue zircon held in secure custody. The project was launched as part of a broader strategy by NYBlue to corner the global blue zircon market, following the company's acquisition of over one million carats of the gemstone.^{[13][14]}

Regulatory framework in the United States

In the United States, the tokenization of real-world assets has become a key area of focus for federal financial policy. According to a 2025 report from the President's Working Group on Digital Asset Markets, the regulatory framework for an RWA is determined not by the blockchain technology used, but by the nature of the underlying asset it represents.^[5] This principle means that if the underlying asset is a security—such as a share in a money market fund or a corporate bond—the resulting token is treated as a "tokenized security" and falls under the jurisdiction of the U.S. Securities and Exchange Commission (SEC), subject to federal securities laws. Conversely, if the token represents a commodity or is used in a derivatives contract, it is subject to the purview of the Commodity Futures Trading Commission (CFTC).^[5] The report emphasizes that establishing a clear taxonomy is essential for the healthy development of the digital asset ecosystem and for investor protection. It contains numerous recommendations for the SEC and CFTC to develop rules that accommodate innovations like decentralized finance (DeFi) and enable the trading of digital assets on federally regulated venues, with the stated goal of positioning the U.S. as a global leader in this field.^[5]

See also

- Asset-backed security
- Blockchain
- Decentralized finance

- Real assets
- Smart contract

External links

- The White House, Digital Assets Report (<https://whitehouse.gov/wp-content/uploads/2025/07/Digital-Assets-Report-EO14178.pdf>)

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